

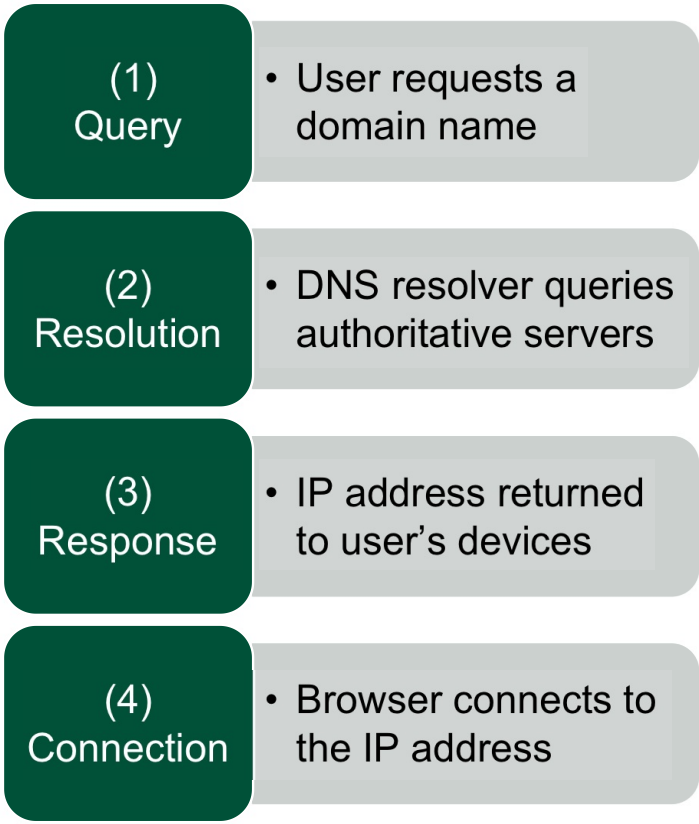

# DNS Cache-Poisoning & Defense via DNSSEC Validation

ISA 656 Network Security

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# What is DNS?

- The internet's phonebook that translates human-readable domain names into IP addresses.
- Example: www.example.com → 10.0.1.20
- Without DNS, users would need to remember IP addresses for every website.



# Real-Life DNS Problems

| | |
|---|---|
| DNS Cache Poisoning | • Attackers inject fake DNS records, redirecting users to malicious sites |
| Man-in-the-Middle | • Intercept and modify DNS queries between user and server |
| DNS Spoofing | • Forged responses trick systems into accepting fake IP Addresses |
| DDoS Attacks | • Amplification attacks overwhelm targets with massive query responses |

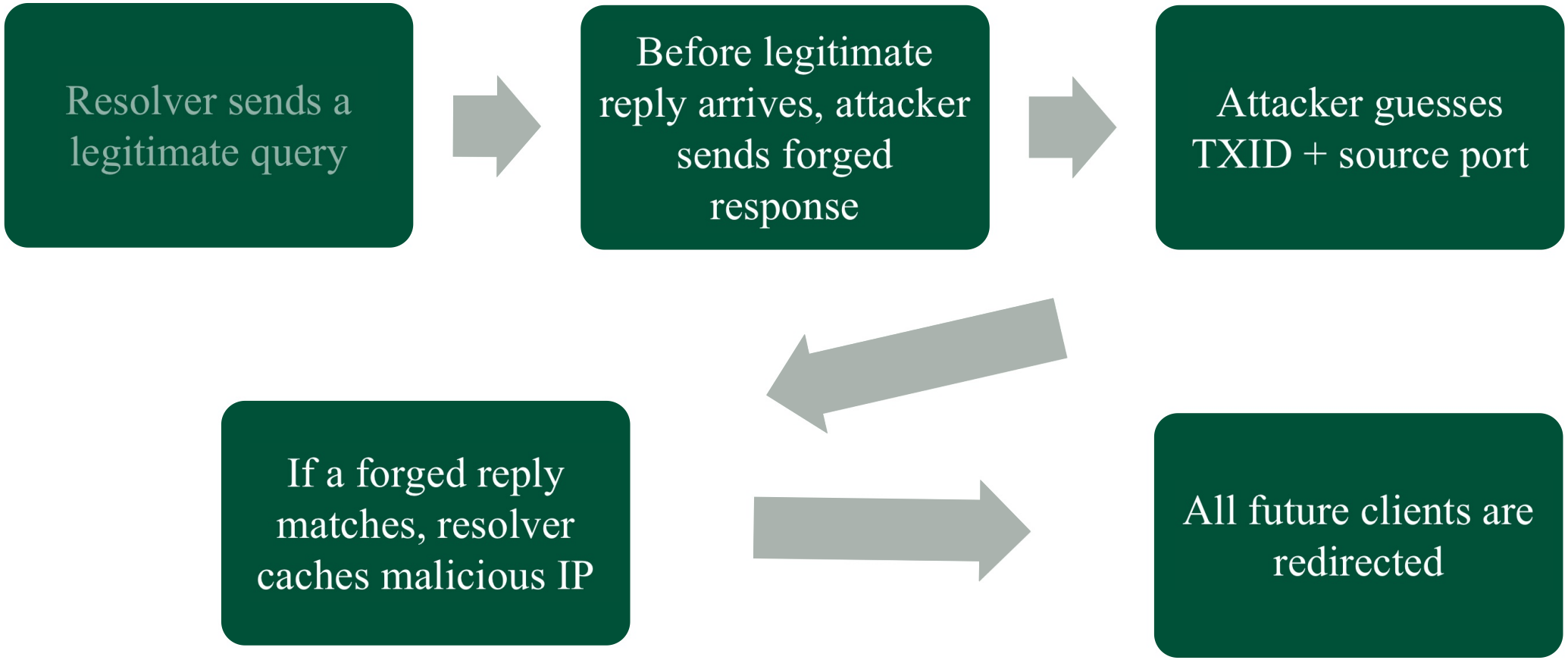


# DNS Cache Poisoning

- DNS resolver caches domain-to-IP mappings to speed up responses.
- Attacker sends forged DNS replies that appear legitimate.
- If the forged reply matches the expected transaction details, the resolver accepts and caches it.
- Future users receive the malicious IP from the poisoned cache.



# DNS Cache Poisoning Attack Workflow

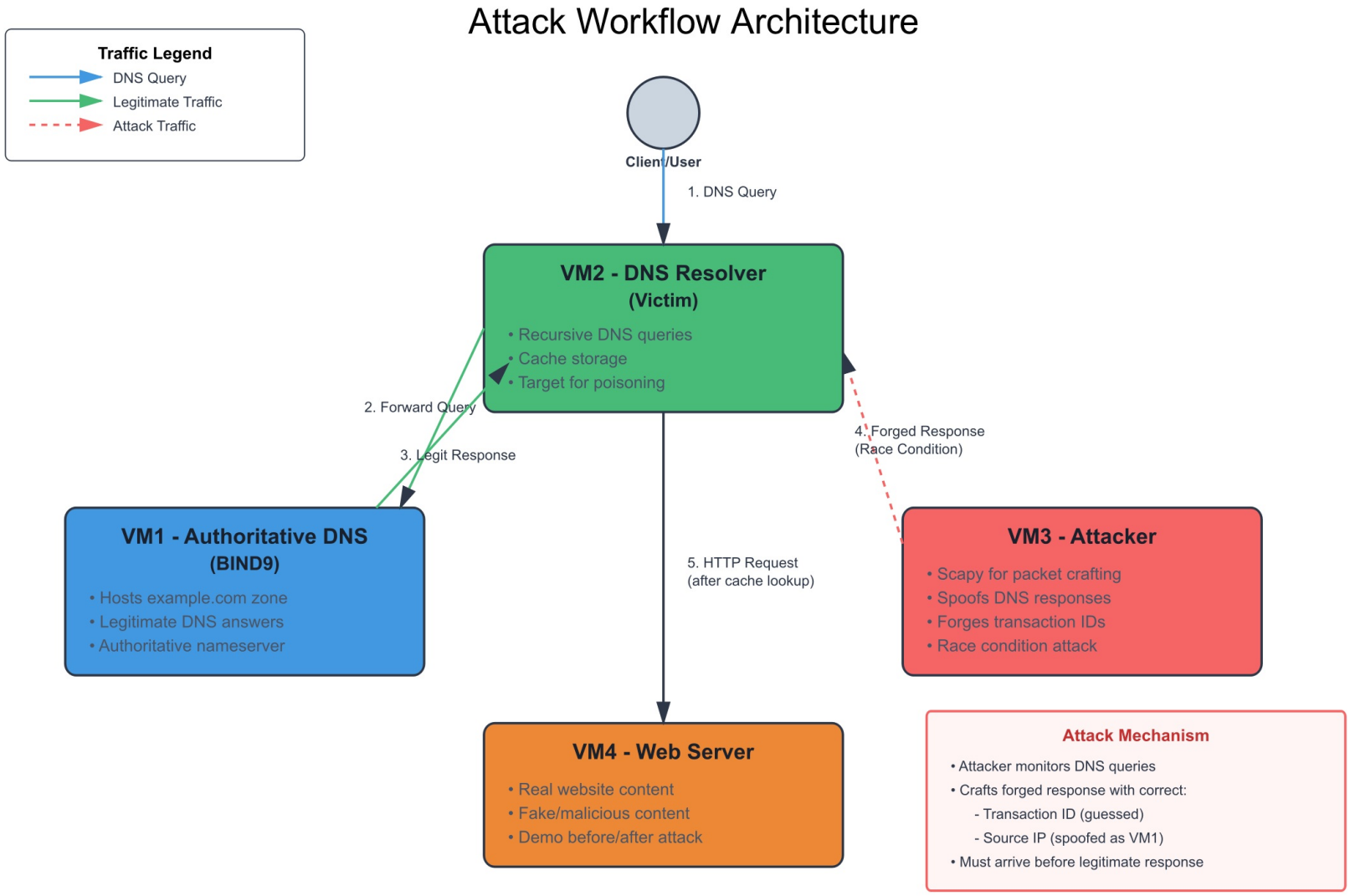


# DNS Cache Poisoning Attack in Real Life Scenario

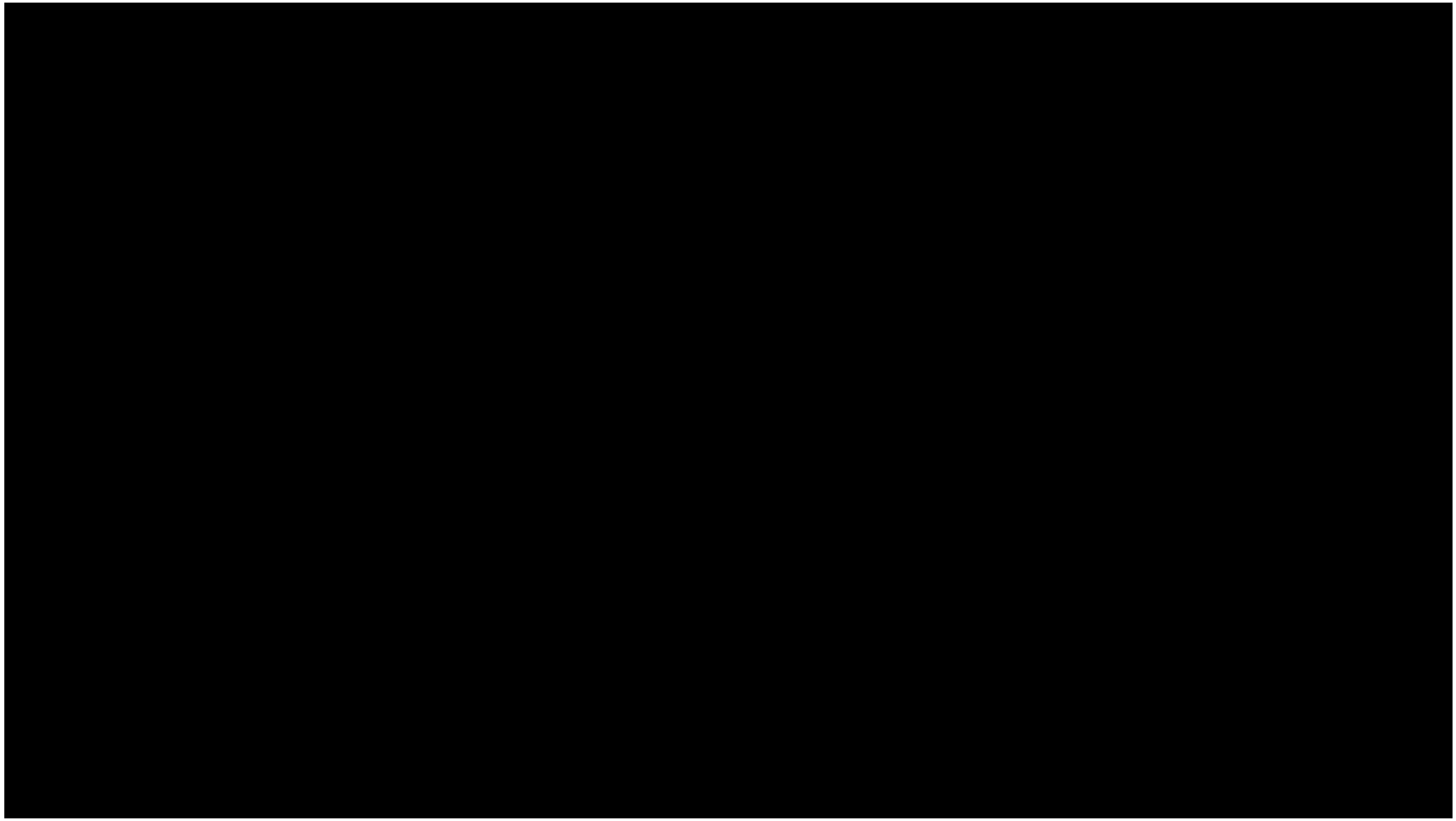
| *Incident* | *Target / Scope* | *Attack Method* | *Impact* | *Impact Sector* | *Significance* |
|---|---|---|---|---|---|
| **2014 Turkey DNS Poisoning** | National-level ISP resolvers | Injected false DNS entries during political unrest | Redirection of social media traffic to government-controlled pages | **Public Communications & Social Media** | Demonstrated use of DNS poisoning for censorship and population-wide manipulation |
| **2010 Brazil ISP Poisoning** | Multiple Brazilian ISPs | Poisoned DNS cache for banking domains, redirecting to phishing sites | Large-scale credential theft and fraudulent financial transactions | **Banking & Financial Services** | Showed financial motivation and massive scalability of poisoning attacks |
| **2019 MyEtherWallet (MEW) Attack** | Cryptocurrency users via Route 53 DNS paths | BGP hijacking enabled DNS poisoning to redirect MEW traffic | Over $150,000 in cryptocurrency stolen | **Cryptocurrency & Blockchain Services** | Proved that even modern cloud DNS and high-value financial platforms can be compromised |



# How DNS Cache Poisoning Works? (Implementation)



# Demo Video



# How can we stop DNS Cache Poisoning

1. **Validate Authenticity of DNS Responses**
   Resolvers should verify that DNS data comes from a legitimate source and has not been altered during transit.
2. **Enforce a Chain of Trust for DNS Records**
   Each DNS record should be verifiable back to a trusted authority so attackers cannot insert unauthorized or fake information.
3. **Allow Only Verified Records to Enter Cache**
   Resolvers should cache only those DNS answers that pass strict validation, preventing poisoned data from being stored or reused.



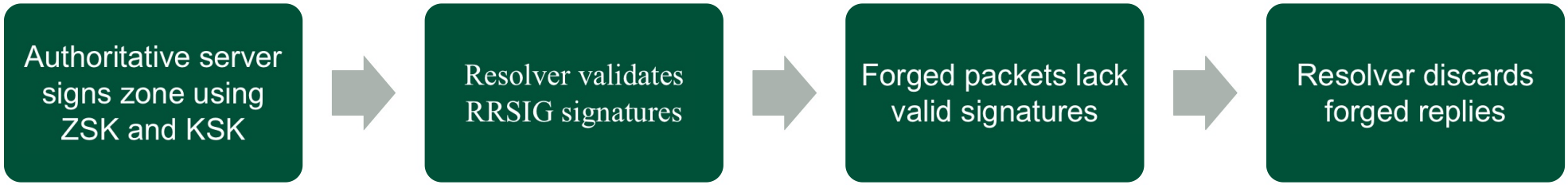
# What is DNSSEC

- DNSSEC adds cryptographic signatures to DNS records to ensure authenticity and integrity.

- Instead of trusting DNS responses blindly, DNSSEC allows validation of every response through a chain of trust.



# DNSSEC Defense Workflow



# End-to-End DNSSEC Workflow

- Start containers
- Start attack (unsigned) → poisoning succeeds
- Enable DNSSEC
- Start attack again → poisoning blocked
- View experimental results



# Attack Blocked (DNSSEC ON)

- **DNSSEC Mode Results:**
- Trials: 100
- Successful poisonings: **0**
- All forged replies rejected
- Resolver logs show *“validation failure*

A1 | fx | trial

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | trial | mode | outcome | blocked_by | duration_sec |
| 2 | 1 | dnssec | blocked | TRUE | 2.9018 |
| 3 | 2 | dnssec | blocked | TRUE | 2.8328 |
| 4 | 3 | dnssec | blocked | TRUE | 2.8366 |
| 5 | 4 | dnssec | blocked | TRUE | 2.8525 |
| 6 | 5 | dnssec | blocked | TRUE | 3.0024 |
| 7 | 6 | dnssec | blocked | TRUE | 2.7781 |
| 8 | 7 | dnssec | blocked | TRUE | 2.8832 |
| 9 | 8 | dnssec | blocked | TRUE | 2.9053 |
| 10 | 9 | dnssec | blocked | TRUE | 2.9655 |





# Conclusion

- DNS cache poisoning is practical and dangerous
- DNSSEC is an effective, cryptographically strong defense
- Slight performance overhead is acceptable
- Our lab demonstrates complete attack → defense → measurement

# References

*[1] O. Alsad and Q. A. Al-Haija, "DNS cache poisoning attack detection: a systematic review," 7th IET Smart Cities Symposium (SCS 2023), Hybrid Conference, Bahrain, 2023, pp. 426-432, doi: 10.1049/icp.2024.0962.*

*[2] S. Abirami and R. Naresh, "DNS Enhancement with DNSSEC and DoT for Enhanced Online Security," 2024 2nd International Conference on Networking and Communications (ICNWC), Chennai, India, 2024, pp. 1-11, doi: 10.1109/ICNWC60771.2024.10537516.*

*[3] N. Alexiou, S. Basagiannis, P. Katsaros, T. Dashpande and S. A. Smolka, "Formal Analysis of the Kaminsky DNS Cache-Poisoning Attack Using Probabilistic Model Checking," 2010 IEEE 12th International Symposium on High Assurance Systems Engineering, San Jose, CA, USA, 2010, pp. 94-103, doi: 10.1109/HASE.2010.25.*

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# Thank You